In [1]:
```
#This notebook is inspired by some notebooks of Eleftherios Garyfallidis ava

# !pip install dipy

%matplotlib inline
```

Connectivity Matrices, ROI Intersections and Density Maps

This example is meant to be an introduction to some of the streamline tools available in DIPY_. Some of the functions covered in this example are `target`, `connectivity_matrix` and `density_map`. `target` allows one to filter streamlines that either pass through or do not pass through some region of the brain, `connectivity_matrix` groups and counts streamlines based on where in the brain they begin and end, and finally, density map counts the number of streamlines that pass through every voxel of some image.

To get started we'll need to have a set of streamlines to work with. We'll use EuDX along with the CsaOdfModel to make some streamlines. Let's import the modules and download the data we'll be using.

This is script is not transparent to you, as it is downloading underneath data. In practice, each time you do a tractography you need 3 files. One Nifti file with all gradient direction scans (a 4D file), a 2 textual files comprising the sensitivity of the DWI sequence (the b-vals), and the associated directions (the b-vecs). Lastly, you would need a T1 volume to create the mask.

You can find the files clearly visible here:
https://www.kaggle.com/code/chrisfilo/diffusion-weighted-imaging-dwi-analysis-with-dipy/input

In [2]:
```
import numpy as np
from scipy.ndimage import binary_dilation

from dipy.core.gradients import gradient_table
from dipy.data import get_fnames
from dipy.io.gradients import read_bvals_bvecs
from dipy.io.image import load_nifti_data, load_nifti, save_nifti
from dipy.direction import peaks
from dipy.reconst import shm
from dipy.tracking import utils
from dipy.tracking.local_tracking import LocalTracking
from dipy.tracking.stopping_criterion import BinaryStoppingCriterion
from dipy.tracking.streamline import Streamlines
```

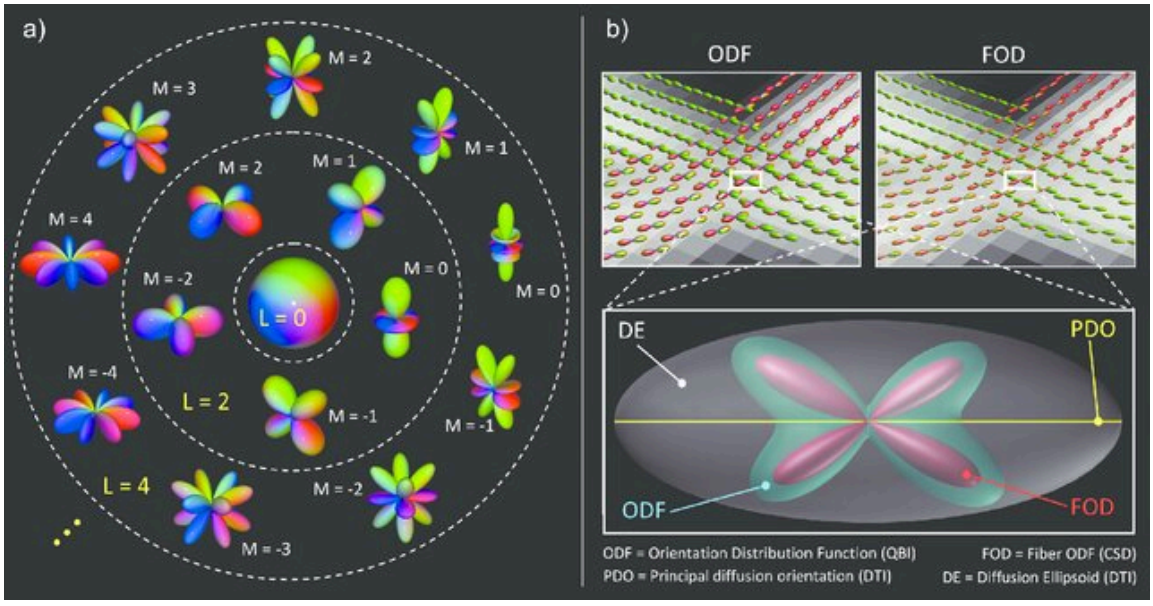
```
hardi_fname, hardi_bval_fname, hardi_bvec_fname = get_fnames('stanford_hardi
label_fname = get_fnames('stanford_labels')
t1_fname = get_fnames('stanford_t1')

data, affine, hardi_img = load_nifti(hardi_fname, return_img=True)
labels = load_nifti_data(label_fname)
t1_data = load_nifti_data(t1_fname)
bvals, bvecs = read_bvals_bvecs(hardi_bval_fname, hardi_bvec_fname)
gtab = gradient_table(bvals, bvecs)
```

```
/Users/antek/Desktop/Studia/MAGISTERKA/2_SEM/machine-learning-applied-to-neu
roimaging/.venv/lib/python3.12/site-packages/tqdm/auto.py:21: TqdmWarning: I
Progress not found. Please update jupyter and ipywidgets. See https://ipywid
gets.readthedocs.io/en/stable/user_install.html
  from .autonotebook import tqdm as notebook_tqdm
[2026-10-24 23:58:26][dipy] INFO: Dataset is already in place. If you want t
o fetch it again please first remove the folder /Users/antek/.dipy/stanford_
hardi
[2026-10-24 23:58:26][dipy] INFO: Dataset is already in place. If you want t
o fetch it again please first remove the folder /Users/antek/.dipy/stanford_
hardi
[2026-10-24 23:58:26][dipy] INFO: Dataset is already in place. If you want t
o fetch it again please first remove the folder /Users/antek/.dipy/stanford_
hardi
```

We've loaded an image called `labels_img` which is a map of tissue types such that every integer value in the array `labels` represents an anatomical structure or tissue type [#]_. For this example, the image was created so that white matter voxels have values of either 1 or 2. We'll use `peaks_from_model` to apply the `CsaOdfModel` to each white matter voxel and estimate fiber orientations which we can use for tracking. We will also dilate this mask by 1 voxel to ensure streamlines reach the grey matter.

In the line "csa_model = CsaOdfModel(gtab, sh_order=6)" sh_order define the "L" as in the figure below:



In [3]:
```
white_matter = binary_dilation((labels == 1) | (labels == 2))
csamodel = shm.CsaOdfModel(gtab, 6)
csapeaks = peaks.peaks_from_model(model=csamodel,
                                  data=data,
                                  sphere=peaks.default_sphere,
                                  relative_peak_threshold=.8,
                                  min_separation_angle=45,
                                  mask=white_matter)
```

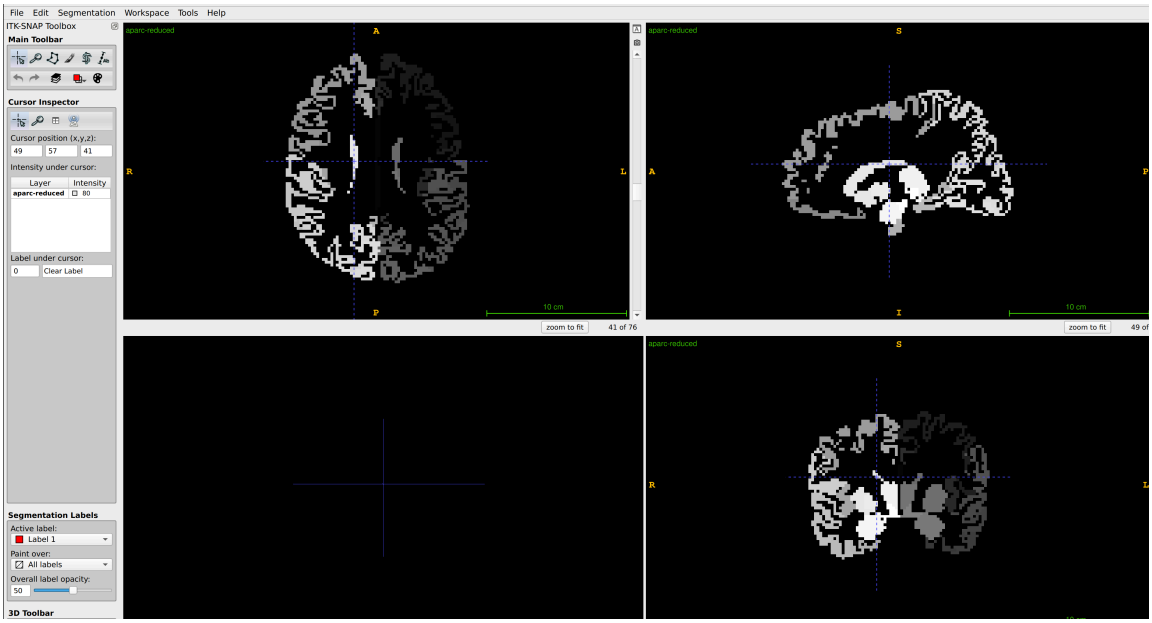
Now we can use EuDX to track all of the white matter. To keep things reasonably fast we use `density=1` which will result in 1 seeds per voxel. The stopping criterion, determining when the tracking stops, is set to stop when the tracking exits the white matter.

In [4]:
```
affine = np.eye(4)
seeds = utils.seeds_from_mask(white_matter, affine, density=1)
stopping_criterion = BinaryStoppingCriterion(white_matter)

streamline_generator = LocalTracking(csapeaks, stopping_criterion, seeds,
                                     affine=affine, step_size=0.5)
streamlines = Streamlines(streamline_generator)
```

The first of the tracking utilities we'll cover here is `target`. This function takes a set of streamlines and a region of interest (ROI) and returns only those streamlines that pass through the ROI. The ROI should be an array such that the voxels that belong to the ROI are `True` and all other voxels are `False` (this type of binary array is sometimes called a mask). This function can also exclude all the streamlines that pass through an ROI by setting the `include` flag to `False`. In this example we'll target the streamlines of the corpus callosum. Our `labels` array has a sagittal slice of the corpus callosum identified by the label value 2. We'll create an ROI mask from that label and create two sets of streamlines, those that intersect with the ROI and those that don't.

To see the labels look at the Nifti file with the labels (aparc-reduced.nii):



In [5]:
```
cc_slice = labels == 2
cc_streamlines = utils.target(streamlines, affine, cc_slice)
cc_streamlines = Streamlines(cc_streamlines)

from dipy.io.stateful_tractogram import Space, StatefulTractogram
from dipy.io.streamline import save_trk

# Save streamlines
sft = StatefulTractogram(cc_streamlines, hardi_img, Space.VOX)
save_trk(sft, "cc_streamlines.trk")



other_streamlines = utils.target(streamlines, affine, cc_slice,
                                 include=False)
other_streamlines = Streamlines(other_streamlines)
assert len(other_streamlines) + len(cc_streamlines) == len(streamlines)
```

Now let's visualize the generated tracts with Trackvis.

Download it here, and keep the license file: https://trackvis.org/download/

If you are running it on Ubuntu 22, you probably will need additional libraries:

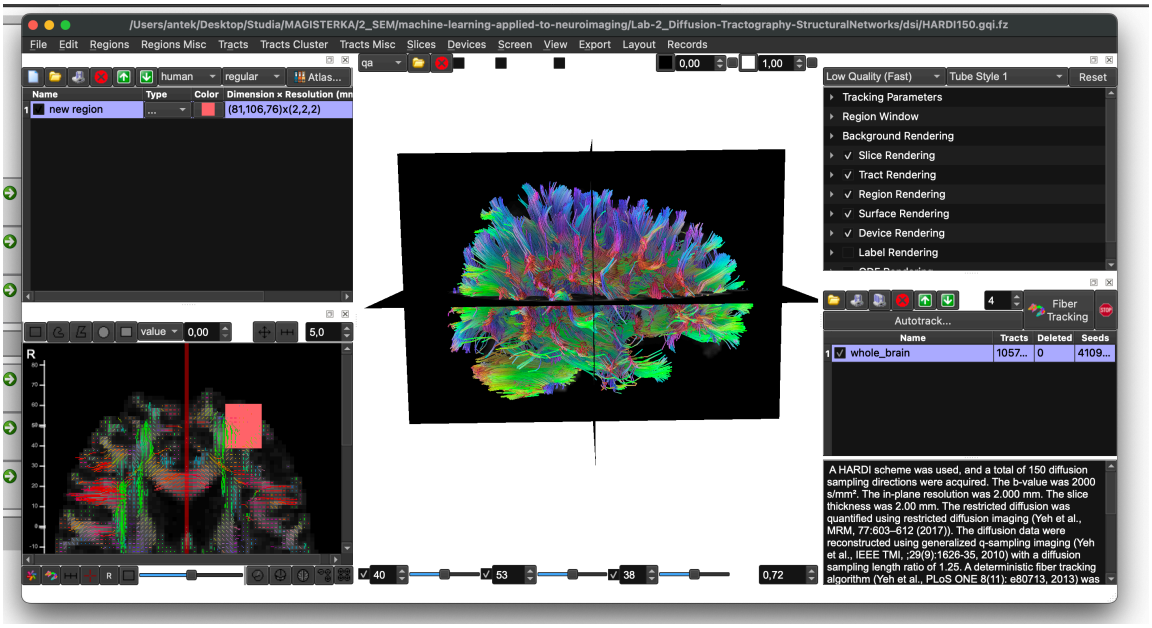
sudo apt-get install libjpeg62

sudo add-apt-repository ppa:linuxuprising/libpng12

sudo apt update

sudo apt install libpng12-0

Whole-brain deterministic tractography of the same Stanford HARDI data (GQI reconstruction), made with DSI Studio 2024



Once we've targeted the corpus callosum ROI, we might want to find out which regions of the brain are connected by these streamlines. To do this we can use the `connectivity_matrix` function. This function takes a set of streamlines and an array of labels as arguments. It returns the number of streamlines that start and end at each pair of labels and it can return the streamlines grouped by their endpoints. Notice that this function only considers the endpoints of each streamline.

In [6]:
```
M, grouping = utils.connectivity_matrix(cc_streamlines, affine,
                                        labels.astype(np.uint8),
                                        return_mapping=True,
                                        mapping_as_streamlines=True)
M[:3, :] = 0
M[:, :3] = 0
```

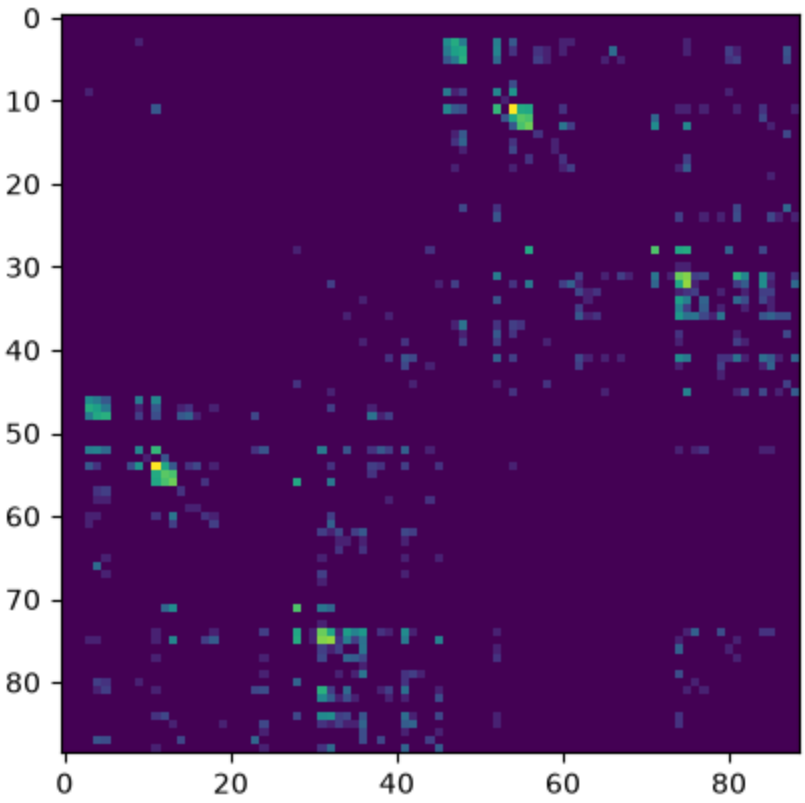
We've set `return_mapping` and `mapping_as_streamlines` to `True` so that `connectivity_matrix` returns all the streamlines in `cc_streamlines` grouped by their endpoint.

Because we're typically only interested in connections between gray matter regions, and because the label 0 represents background and the labels 1 and 2 represent white matter, we discard the first three rows and columns of the connectivity matrix.

We can now display this matrix using matplotlib. We display it using a log scale to make small values in the matrix easier to see.

In [7]:
```
import matplotlib.pyplot as plt
```

```
plt.imshow(np.log1p(M), interpolation='nearest')
plt.savefig("connectivity.png")
```

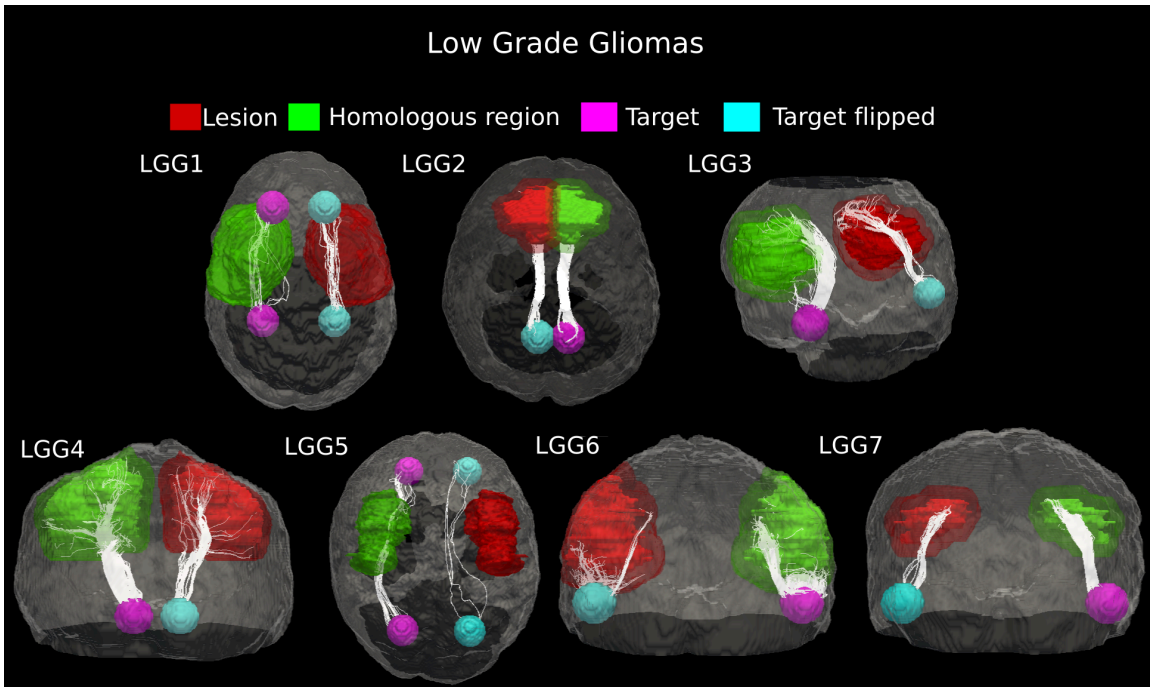


In our example track there are more streamlines connecting regions 11 and 54 than any other pair of regions. These labels represent the left and right superior frontal gyrus respectively. These two regions are large, close together, have lots of corpus callosum fibers and are easy to track so this result should not be a surprise to anyone.

However, the interpretation of streamline counts can be tricky. The relationship between the underlying biology and the streamline counts will depend on several factors, including how the tracking was done, and the correct way to interpret these kinds of connectivity matrices is still an open question in the diffusion imaging literature.

The next function we'll demonstrate is `density_map`. This function allows one to represent the spatial distribution of a track by counting the density of streamlines in each voxel. For example, let's take the track connecting the left and right superior frontal gyrus.

This is the screenshot from another example from Campanella et al. 2014



In [8]:
```
lr_superiorfrontal_track = grouping[11, 54]
shape = labels.shape
dm = utils.density_map(lr_superiorfrontal_track, affine, shape)
```

Let's save this density map and the streamlines so that they can be visualized together. In order to save the streamlines in a ".trk" file we'll need to move them to "trackvis space", or the representation of streamlines specified by the trackvis Track File format.

In [9]:
```
from dipy.io.stateful_tractogram import Space, StatefulTractogram
from dipy.io.streamline import save_trk

# Save density map
save_nifti("lr-superiorfrontal-dm.nii.gz", dm.astype("int16"), affine)

lr_sf_trk = Streamlines(lr_superiorfrontal_track)

# Save streamlines
sft = StatefulTractogram(lr_sf_trk, hardi_img, Space.VOX)
save_trk(sft, "lr-superiorfrontal.trk")
```

.. rubric:: Footnotes

.. [#] The image `aparc-reduced.nii.gz`, which we load as `labels_img`, is a modified version of label map `aparc+aseg.mgz` created by FreeSurfer. The corpus callosum region is a combination of the FreeSurfer labels 251-255. The remaining FreeSurfer labels were re-mapped and reduced so that they lie between 0 and 88. To see the FreeSurfer region, label and name, represented by each value, see `label_info.txt` in `~/.dipy/stanford_hardi`. .. [#] An affine transformation is a mapping between two coordinate systems that can represent scaling, rotation, shear,

translation and reflection. Affine transformations are often represented using a 4x4 matrix where the last row of the matrix is `[0, 0, 0, 1]`.